MICHAEL AKOREDE FATAYO

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RESEARCH INTERESTS

Real-time machine learning systems; fraud and anomaly detection; imbalanced classification; causal feature engineering and leakage-free evaluation protocols; streaming data architectures; trustworthy and explainable machine learning for financial systems.

EDUCATION

Master of Science (MSc), Computer Science Lead City University, Ibadan, Nigeria

Expected 2026. Thesis: "FraudShield: A Real-Time Machine Learning Framework for Fraud Detection in High-Volume Financial Transaction Streams."

Bachelor of Science (BSc), Computer Science Imo State University, Owerri, Nigeria

2020 – 2024. First Class Honours; GPA 4.27/5.00.

RESEARCH EXPERIENCE

FraudShield: Real-Time Fraud Detection Research Lead City University

MSc Thesis Research; independent researcher | Code: github.com/mafatayo/fraudshield

- Designed and implemented a real-time streaming fraud-detection architecture (Apache Kafka, Spark Streaming, Redis, XGBoost, PostgreSQL), measuring a median scoring latency of 0.21ms and batched throughput of 1.85 million transactions/second on commodity hardware, against an explicit one-second end-to-end latency budget.
- Engineered a 23-feature causal scheme across three feature families (transaction, balance-consistency, velocity), constructed to eliminate look-ahead leakage, and validated it under a strict temporal train/test protocol at full dataset scale (6,362,620 transactions).
- Evaluated four detectors (rule-based baseline, logistic regression, random forest, XGBoost) under both a conventional random split and a stricter temporal split; XGBoost achieved PR-AUC 0.9997 under the temporal protocol against a rule-based baseline recall of 0.19%.
- Ran a feature-group ablation study (five nested feature sets) to separate genuine learned signal from a dataset-specific artifact in the balance-consistency features, and to quantify the independent contribution of streaming-derived velocity features — reframing the evaluation methodology as a contribution in its own right, not only the model's accuracy.

Software Quality Assurance Process: A Consolidated Definition and a Lifecycle-Mapped Taxonomy of Tool Support Lead City University

Independent research paper, SEN 712 (Software Quality Assurance)

- Conducted a structured narrative review across IEEE Std 730/1028/829, ISO/IEC 25010, and the SWEBOK Guide to resolve persistent terminological conflation between quality assurance, quality control, and testing, producing a consolidated working definition of the SQA process.
- Constructed a lifecycle-mapped taxonomy of QA tool categories (requirements traceability, static analysis, review support, testing frameworks, CI, defect tracking) to replace vendor-driven tool selection with a lifecycle-stage-driven rationale; proposed empirical validation of the taxonomy against the FraudShield pipeline as future work.

PUBLICATIONS & PRESENTATIONS

Fatayo, M. A. "FraudShield: A Real-Time Machine Learning Framework for Fraud Detection in High-Volume Financial Transaction Streams." Manuscript prepared for submission, Bridges (5th International Conference on Frontiers in Multidisciplinary Science and Artificial Intelligence, FASCON 2026), Lead City University, Ibadan, Nov. 2026.

Conference abstract submitted, FASCON 2026 (Faculty of Natural and Applied Sciences, Lead City University), Ibadan, Nigeria, Nov. 2026.

Fatayo, M. A. “Software Quality Assurance Process: A Consolidated Definition and a Lifecycle-Mapped Taxonomy of Tool Support.” Preprint, Zenodo, Sept. 2026. https://doi.org/10.5281/zenodo.22341782

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, SQL.

Machine Learning & Data: pandas, scikit-learn, XGBoost, imbalanced-classification methods, causal feature engineering, model evaluation (PR-AUC, ROC-AUC).

Systems: Apache Kafka, Spark Streaming, Redis, PostgreSQL, streaming architecture design and latency benchmarking.

Full-Stack Development: React, Node.js, MongoDB, Socket.io.

PROFESSIONAL EXPERIENCE

Founder & Lead Developer, Khor
Built a full-stack SaaS platform for gym management (React, Node.js, MongoDB, Socket.io) serving fitness businesses across Africa, including a real-time QR check-in system and live revenue-tracking dashboards.

Chief Technology Officer, Khor Consolidated Ile-Ife, Osun State | Sept 2020 – Dec 2023
Managed technical infrastructure and content platform operations.

Front End Developer, Upwork Global Inc. Remote | Apr 2021 – Sept 2025
Developed responsive, browser-compatible web applications and e-commerce platforms for international clients.

CERTIFICATIONS & HONORS

Software Engineering Certification, ALX Africa (2023).

AI Programming with Python Nanodegree, Udacity (2024).

First Class Honours, BSc Computer Science, Imo State University (GPA 4.27/5.00).